Class 10: Structural Bioinformatics (Pt. 1)

Dea Sinaga (PID: A17725676)

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Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB).

PDB statistics

What is in the PDB in terms of molecule type and structure determination method?

Read a CSV file of current PDB stats obtained from https://www.rcsb.org/stats/summary

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

```
           Molecular.Type   X.ray     EM    NMR Integrative Multiple.methods
1          Protein (only) 180,758 23,111 12,813         348              229
2 Protein/Oligosaccharide  10,488  3,741     34           8               11
3              Protein/NA   9,205  6,751    287          26                8
4     Nucleic acid (only)   3,154    250  1,578           3               15
5                   Other     178     27     35           4                0
```

```
6  Oligosaccharide (only)      11       0       6             0                 1
  Neutron Other   Total
1      84     32 217,375
2       1      0  14,283
3       0      0  16,277
4       3      1   5,004
5       0      0     244
6       0      4      22
```

> **Q1**: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```r
pdb$X.ray
```

```
[1] "180,758" "10,488"  "9,205"   "3,154"   "178"     "11"
```

This print out above 'pdb$X.ray' is "character", not "numeric". Therefore, I can't do math with it. We need to fix this…

Two functions that can help here are 'sub()' and

```r
# We want to get rid (or sub out) commas:
x <- pdb$X.ray
tmp <- sub(",", "", x)
sum(as.numeric(tmp))
```

```
[1] 203794
```

We could make a function to do this:

```r
rm.comma <- function(x) {
  tmp <- sub(",", "", x)
  sum(as.numeric(tmp))
}
```

```r
rm.comma(pdb$'X-ray')
```

```
[1] 0
```

```r
rm.comma(pdb$EM)
```

```
[1] 33880
```

We could also use a different input function for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file)

```r
library(readr)

read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): Molecular Type
dbl (4): Integrative, Multiple methods, Neutron, Other
num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# A tibble: 6 x 9
  `Molecular Type`    `X-ray`    EM   NMR Integrative `Multiple methods` Neutron
  <chr>                 <dbl> <dbl> <dbl>       <dbl>              <dbl>   <dbl>
1 Protein (only)       180758 23111 12813         348                229      84
2 Protein/Oligosacch~   10488  3741    34           8                 11       1
3 Protein/NA             9205  6751   287          26                  8       0
4 Nucleic acid (only)    3154   250  1578           3                 15       3
5 Other                   178    27    35           4                  0       0
6 Oligosaccharide (o~      11     0     6           0                  1       0
# i 2 more variables: Other <dbl>, Total <dbl>
```

```r
n.tot <- rm.comma(pdb$Total)
n.xray <- rm.comma(pdb$'X-ray')
n.em <- rm.comma(pdb$EM)

n.xray / n.tot * 100
```

```
[1] 0
```

```r
n.em / n.tot * 100
```

```
[1] 13.38046
```

> How many total protein structures are there

```r
pdb$Total[1]
```

```
[1] "217,375"
```

The total number of protein sequences in UniProt is 202,556,314.

```r
217375/202556314 * 100
```

```
[1] 0.1073158
```

> **Key point**: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray).

Visualizing PDB data with Mol-star

Main stand alone web version with all features is at https://molstar.org/viewer/

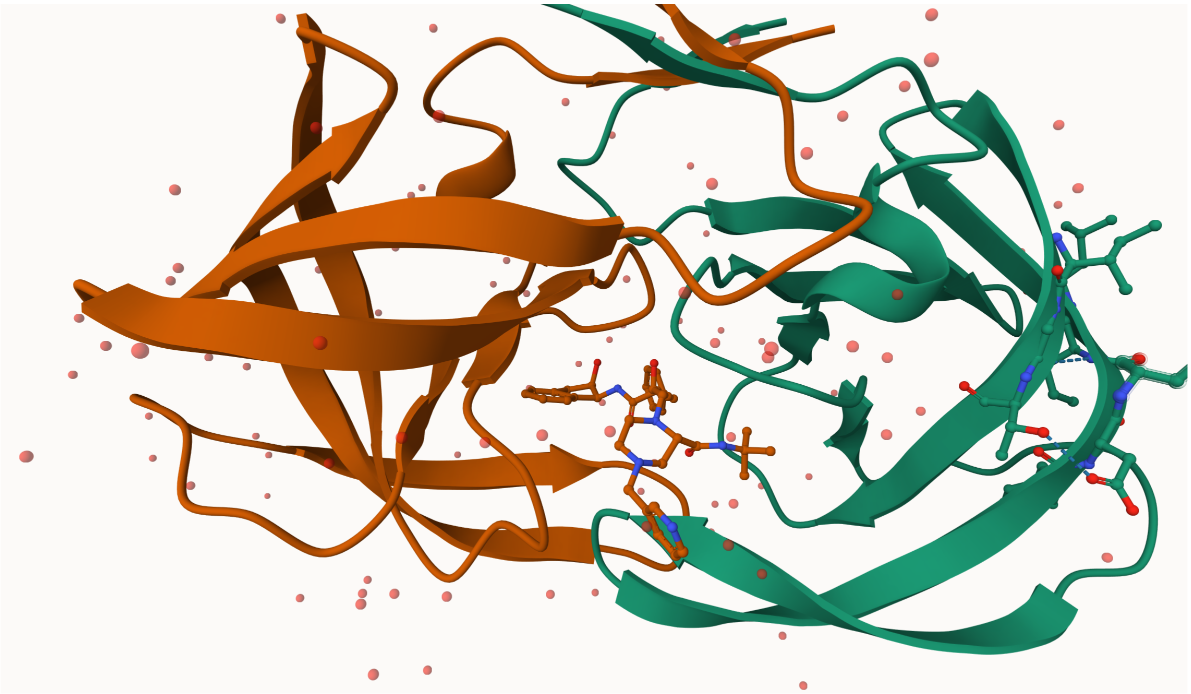
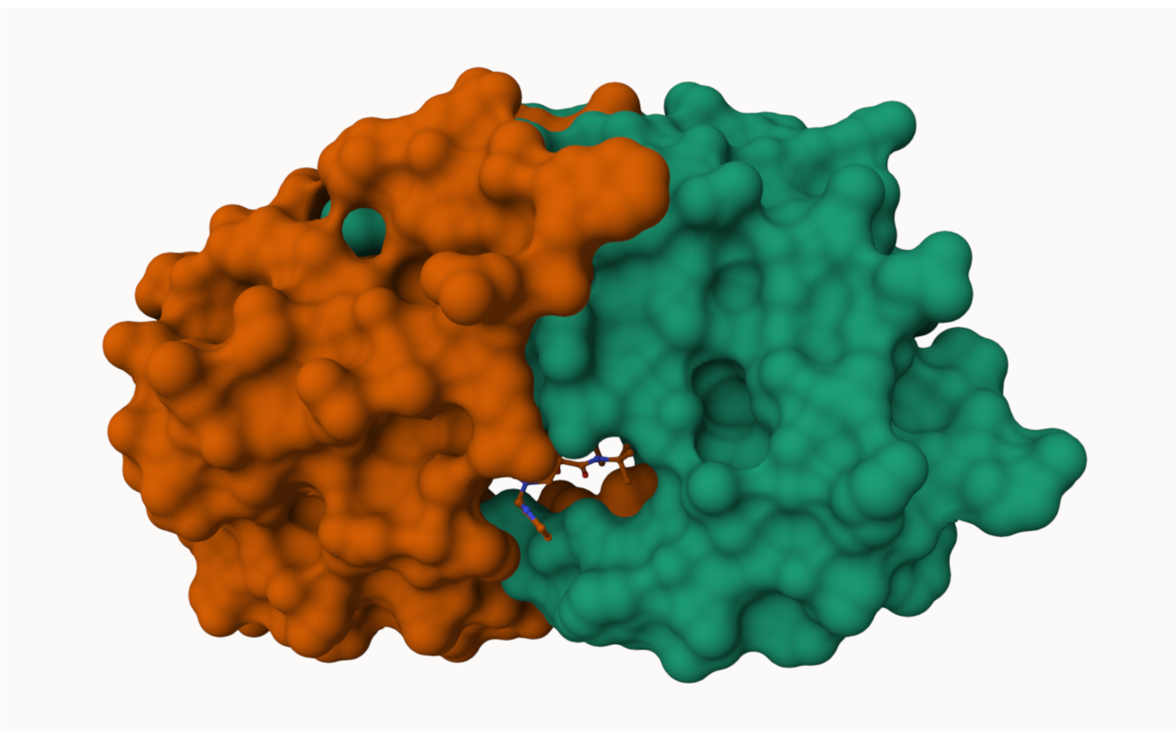
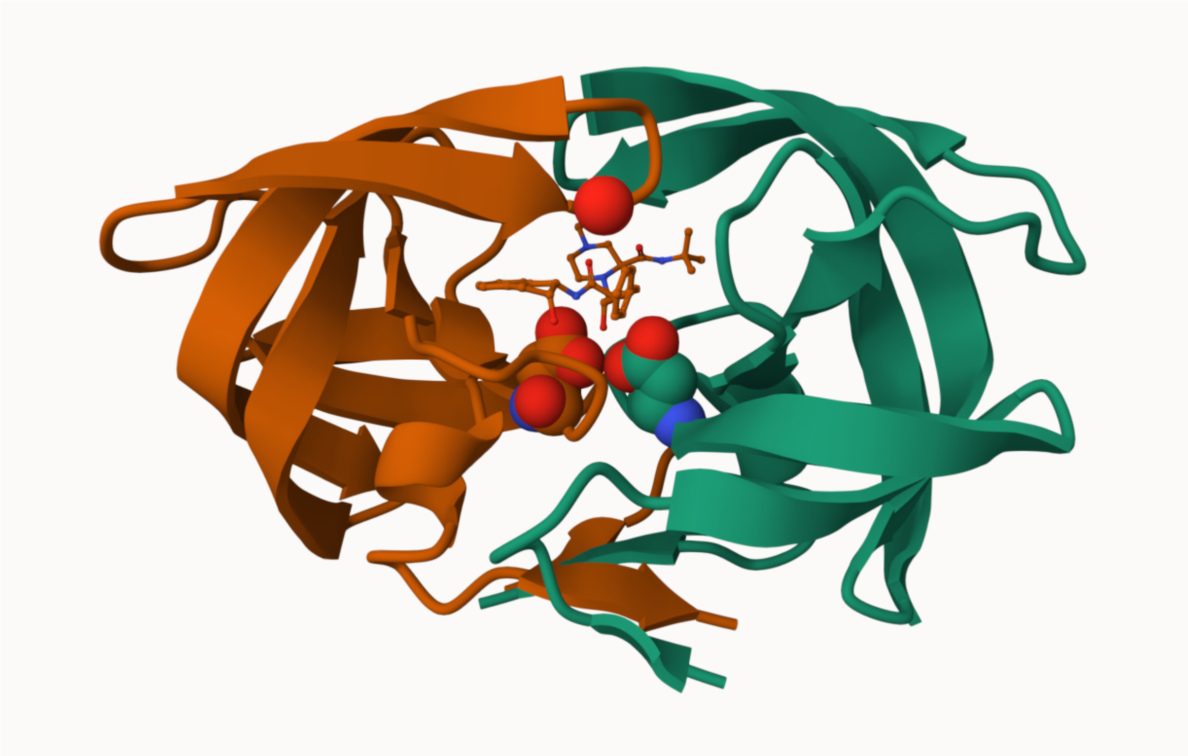


Figure 1: The HIV-protease enzyme is a homodimer with two chains.





Getting started with the Bio3D package

Bio3D is an R package fro CRAN for structural bioinformatics

```r
library(bio3d)

pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
pdb
```

```

 Call:  read.pdb(file = "1hsg")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)
```

```
    Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
    Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

    Non-protein/nucleic Atoms#: 172  (residues: 128)
    Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

  Protein sequence:
     PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
     QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
     ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
     VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
head(pdb$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

There are lots of functions that can work with these ‘pdb’ objects:

```r
head(pdbseq(pdb))
```

```
  1   2   3   4   5   6
"P" "Q" "I" "T" "L" "W"
```

We can have a quick interactive view of any of these ‘pdb’ objects:

```r
library(bio3dview)

view.pdb(pdb)
```

Let's try a custom view.

```r
view.pdb(pdb, colorScheme="sse", backgroundColor="black")
```

> Q. Create a custom view of HIV-Pr highlighting the active site ASP ('resno=25'), the two chains (in your choice of colors), and the ligand all on a custom color background.

```r
active.site <- atom.select(pdb, resno=25)

library(NGLVieweR)

view.pdb(pdb,
         cols <- c("red", "blue"),
         highlight = active.site,
         backgroundColor = "pink",
         highlight.style = "spacefill") |>
  setRock()
```

Predict the flexibility of a given structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a given 'pdb' object:

```r
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

A quick structure summary

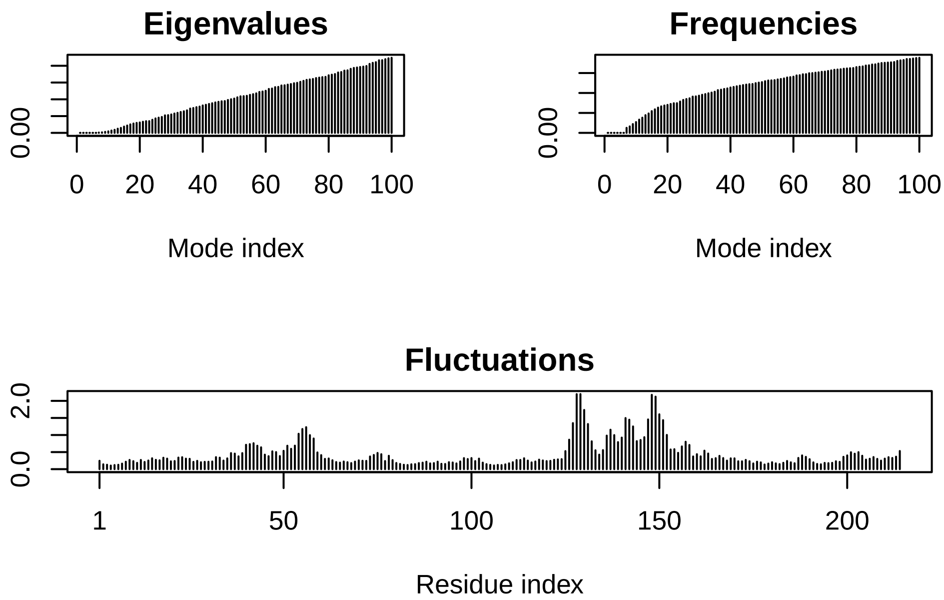
```r
adk
```

```
 Call:  read.pdb(file = "6s36")

   Total Models#: 1
     Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)

     Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 244  (residues: 244)
     Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]

   Protein sequence:
      MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
      VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.009 seconds.
 Diagonalizing Hessian...   Done in 0.179 seconds.
```

```r
plot(m)
```




```r
view.nma(m)
```

Write out the results for viewing in Mol-star:

```r
mktrj(m, file="nma.pdb")
```

Comparative analysis of the ADK family

Our first step is to find a sequence for this family. We will use the database ID "1ake_A" here:

```r
id <- "1ake_A"

aa <- get.seq(id)
```

```
Warning in get.seq(id): Removing existing file: seqs.fasta
```

```
Fetching... Please wait. Done.
```

```r
aa
```

```
              1        .         .         .         .         .         60 
pdb|1AKE|A    MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
              1        .         .         .         .         .         60 

             61        .         .         .         .         .         120 
pdb|1AKE|A    DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
             61        .         .         .         .         .         120 

            121        .         .         .         .         .         180 
pdb|1AKE|A    VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
            121        .         .         .         .         .         180 

            181        .         .         .   214 
pdb|1AKE|A    YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
            181        .         .         .   214 

Call:
  read.fasta(file = outfile)

Class:
  fasta

Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call
```

Search for related sequences in the database

```r
blast <- blast.pdb(aa)
```

```
 Searching ... please wait (updates every 5 seconds) RID = ZZGD7FMF014 
 .........
 Reporting 96 hits
```

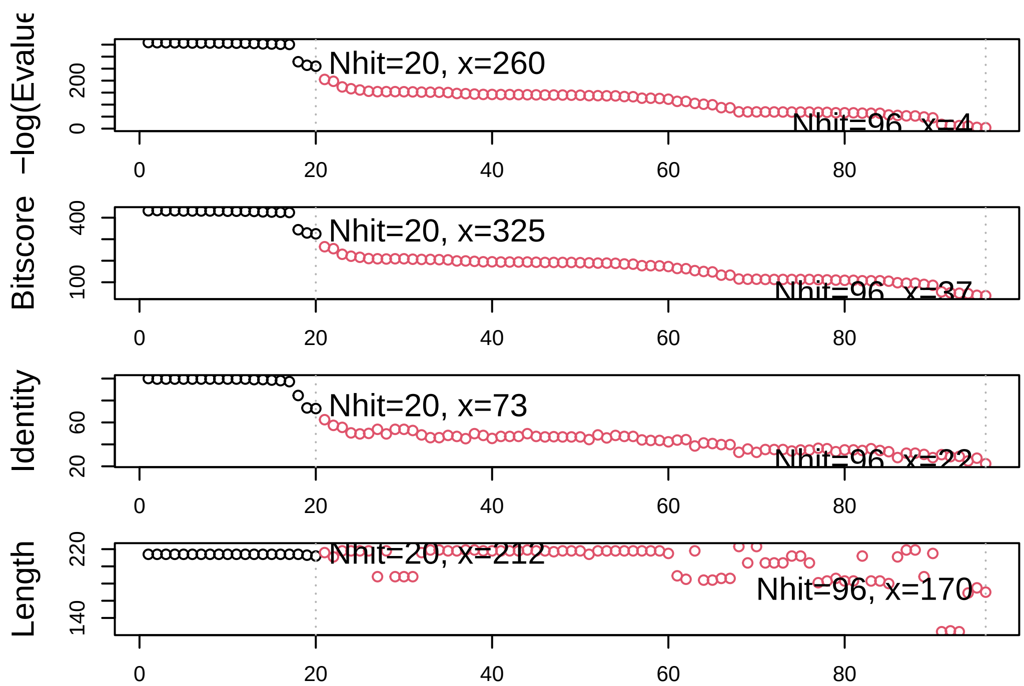
```r
head(blast$hit.tbl)
```

```
       queryid subjectids identity alignmentlength mismatches gapopens q.start
```

```
1 Query_798705      1AKE_A  100.000              214          0        0        1
2 Query_798705      8BQF_A   99.533              214          1        0        1
3 Query_798705      4X8M_A   99.533              214          1        0        1
4 Query_798705      6S36_A   99.533              214          1        0        1
5 Query_798705      9R6U_A   99.533              214          1        0        1
6 Query_798705      9R71_A   99.533              214          1        0        1
  q.end s.start s.end     evalue bitscore positives mlog.evalue pdb.id    acc
1   214       1   214 1.82e-156      432    100.00    358.6044 1AKE_A 1AKE_A
2   214      21   234 2.97e-156      433    100.00    358.1147 8BQF_A 8BQF_A
3   214       1   214 3.25e-156      432    100.00    358.0246 4X8M_A 4X8M_A
4   214       1   214 4.77e-156      432    100.00    357.6409 6S36_A 6S36_A
5   214       1   214 1.06e-155      431     99.53    356.8424 9R6U_A 9R6U_A
6   214       1   214 1.25e-155      431     99.53    356.6775 9R71_A 9R71_A
```

```
hits <- plot(blast)
```

```
  * Possible cutoff values:    260 3
            Yielding Nhits:    20 96

  * Chosen cutoff value of:    260
            Yielding Nhits:    20
```



```r
head(hits$pdb.id)
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A"
```

```r
files <- get.pdb(hits$pdb.id, path="pdbs")
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/1AKE.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/8BQF.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/4X8M.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/6S36.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/9R6U.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/9R71.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/8Q2B.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/8RJ9.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/6RZE.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/4X8H.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/3HPR.pdb exists. Skipping
download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/1E4V.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/5EJE.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/1E4Y.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/3X2S.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/6HAP.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/6HAM.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/8PVW.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/4K46.pdb exists. Skipping
download

Warning in get.pdb(hits$pdb.id, path = "pdbs"): pdbs/4NP6.pdb exists. Skipping
download
```

Align and superpose all these ADK structures

```r
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

```
Reading PDB files:
pdbs/1AKE.pdb
pdbs/8BQF.pdb
pdbs/4X8M.pdb
pdbs/6S36.pdb
pdbs/9R6U.pdb
pdbs/9R71.pdb
pdbs/8Q2B.pdb
```

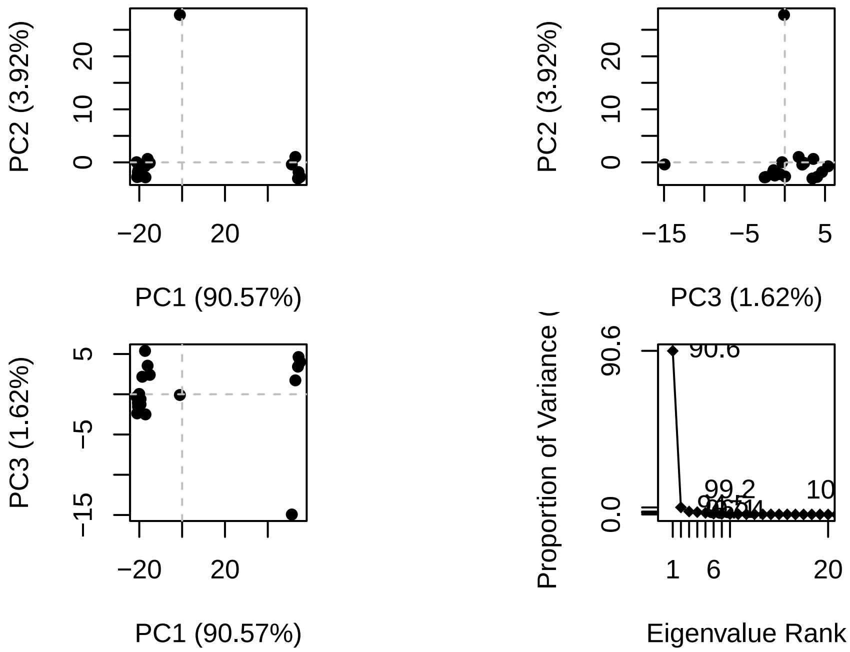
```
pdbs/8RJ9.pdb
pdbs/6RZE.pdb
pdbs/4X8H.pdb
pdbs/3HPR.pdb
pdbs/1E4V.pdb
pdbs/5EJE.pdb
pdbs/1E4Y.pdb
pdbs/3X2S.pdb
pdbs/6HAP.pdb
pdbs/6HAM.pdb
pdbs/8PVW.pdb
pdbs/4K46.pdb
pdbs/4NP6.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
....   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.

Extracting sequences

pdb/seq: 1   name: pdbs/1AKE.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/8BQF.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/4X8M.pdb
pdb/seq: 4   name: pdbs/6S36.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/9R6U.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbs/9R71.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/8Q2B.pdb
```

```
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbs/8RJ9.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbs/6RZE.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10   name: pdbs/4X8H.pdb
pdb/seq: 11   name: pdbs/3HPR.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12   name: pdbs/1E4V.pdb
pdb/seq: 13   name: pdbs/5EJE.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14   name: pdbs/1E4Y.pdb
pdb/seq: 15   name: pdbs/3X2S.pdb
pdb/seq: 16   name: pdbs/6HAP.pdb
pdb/seq: 17   name: pdbs/6HAM.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18   name: pdbs/8PVW.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19   name: pdbs/4K46.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 20   name: pdbs/4NP6.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
```

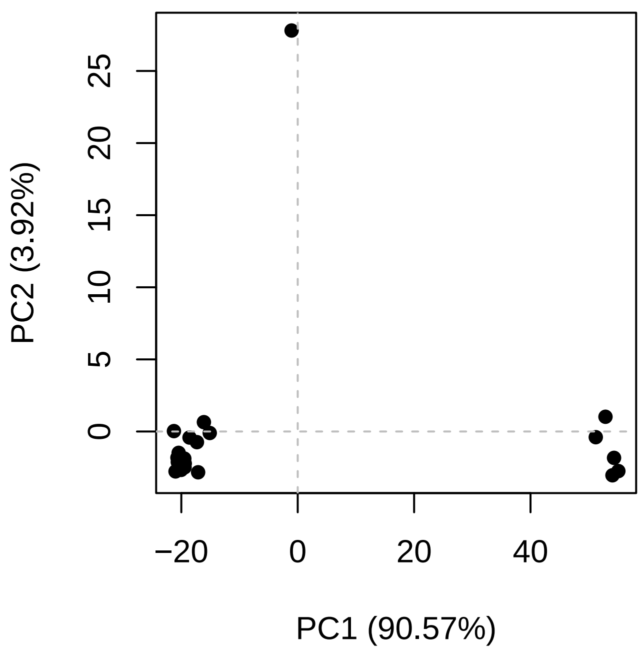
```r
view.pdbs(pdbs)
```

PCA of all this strucural data

```r
pc <- pca(pdbs)
plot(pc)
```




```r
plot(pc, 1:2)
```



Interactive view of the PC1 captured structural differences:

```r
view.pca(pc)
```

```r
mktrj(pc, file = "pca.pdb")
```